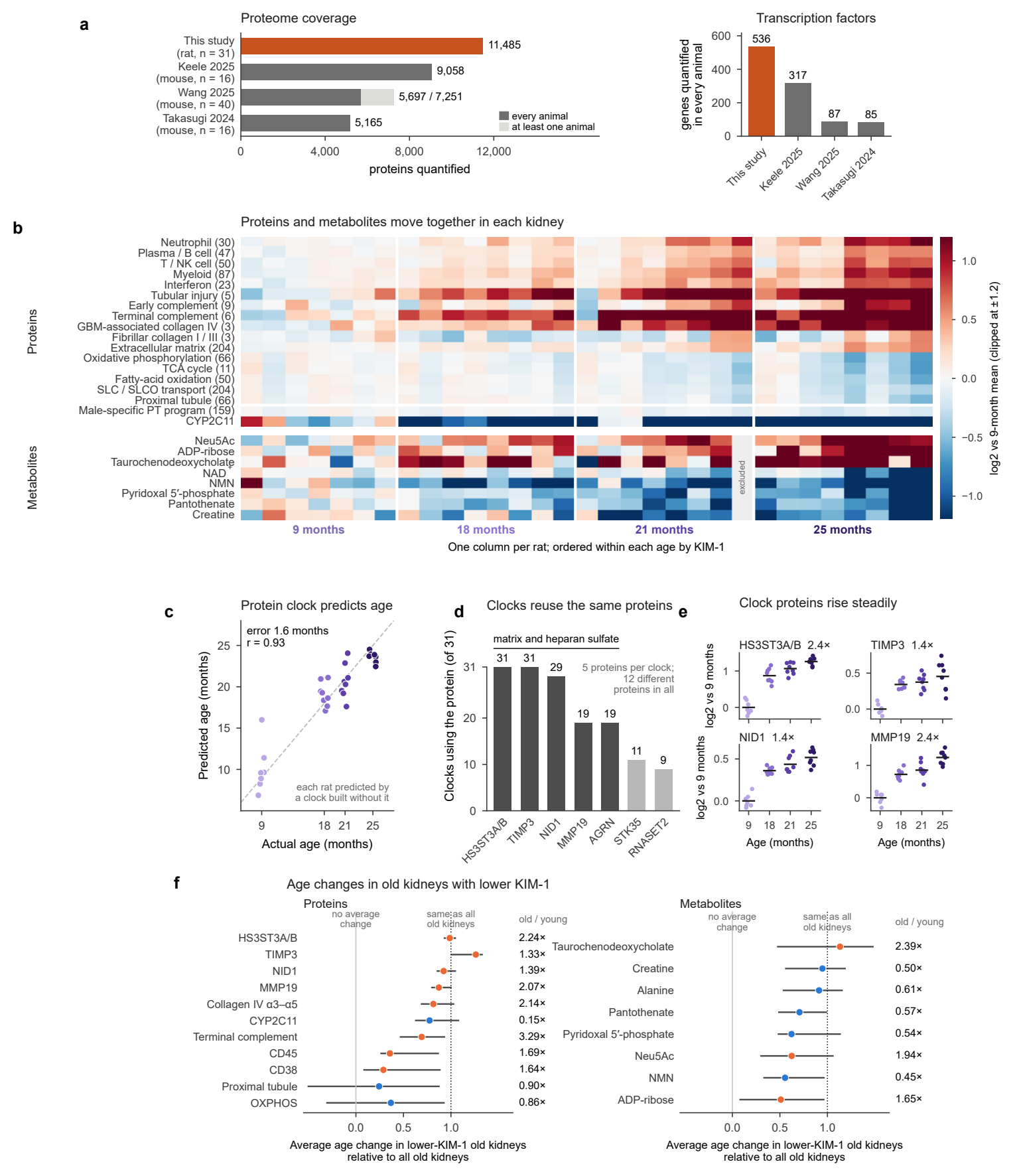

a
Proteome coverage
This study (rat, n = 31)
Keele 2025 (mouse, n = 16)
Wang 2025 (mouse, n = 40)
Takasugi 2024 (mouse, n = 16)
11,485
9,058
5,697 / 7,251
5,165
every animal
at least one animal
0
4,000
8,000
12,000
proteins quantified
Transcription factors
genes quantified in every animal
600
400
200
0
536
317
87
85
This study
Keele 2025
Wang 2025
Takasugi 2024
b
Proteins and metabolites move together in each kidney
Proteins
Neutrophil (30)
Plasma / B cell (47)
T / NK cell (50)
Myeloid (87)
Interferon (23)
Tubular injury (5)
Early complement (9)
Terminal complement (6)
GBM-associated collagen IV (3)
Fibrillar collagen I / III (3)
Extracellular matrix (204)
Oxidative phosphorylation (66)
TCA cycle (11)
Fatty-acid oxidation (50)
SLC / SLCO transport (204)
Proximal tubule (66)
Male-specific PT program (159)
CYP2C11
Metabolites
Neu5Ac
ADP-ribose
Taurochenodeoxycholate
NAD+
NMN
Pyridoxal 5′-phosphate
Pantothenate
Creatine
excluded
9 months
18 months
21 months
25 months
One column per rat; ordered within each age by KIM-1
log2 vs 9-month mean (clipped at ±1.2)
1.0
0.5
0.0
−0.5
−1.0
c
Protein clock predicts age
error 1.6 months
r = 0.93
each rat predicted by a clock built without it
Predicted age (months)
Actual age (months)
d
Clocks reuse the same proteins
matrix and heparan sulfate
5 proteins per clock; 12 different proteins in all
Clocks using the protein (of 31)
31
31
29
19
19
11
9
HS3ST3A/B
TIMP3
NID1
MMP19
AGRN
STK35
RNASET2
e
Clock proteins rise steadily
HS3ST3A/B 2.4×
TIMP3 1.4×
NID1 1.4×
MMP19 2.4×
log2 vs 9 months
Age (months)
f
Age changes in old kidneys with lower KIM-1
Proteins
no average change
same as all old kidneys
old / young
HS3ST3A/B 2.24×
TIMP3 1.33×
NID1 1.39×
MMP19 2.07×
Collagen IV α3–α5 2.14×
CYP2C11 0.15×
Terminal complement 3.29×
CD45 1.69×
CD38 1.64×
Proximal tubule 0.90×
OXPHOS 0.86×
Average age change in lower-KIM-1 old kidneys relative to all old kidneys
Metabolites
Taurochenodeoxycholate 2.39×
Creatine 0.50×
Alanine 0.61×
Pantothenate 0.57×
Pyridoxal 5′-phosphate 0.54×
Neu5Ac 1.94×
NMN 0.45×
ADP-ribose 1.65×